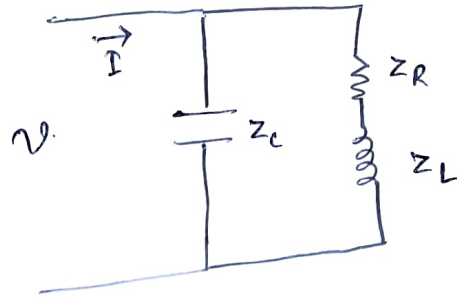


① Given $v = 12\sqrt{2} \cos 5000t$ V

so, $v = 12\sqrt{2}$ V (rms)

$\omega = 5000$ rad/s



$R = 20 \Omega$ (Given)

so, $Z_R = 20 \Omega$

$C = 10 \mu\text{F}$ (Given)

$Z_C = \frac{1}{j\omega C} = -20j \Omega$

$L = 3 \text{ mH}$ (Given)

$Z_L = j\omega L = j15 \Omega$

Now,

$Z_{eq} = (Z_C) \parallel (Z_R + Z_L)$
 $= (-20j) \parallel (20 + j15)$

$Z_{eq} = 24.25 \angle -39.09^\circ \Omega$ (in polar form)

or, $Z_{eq} = (18.82 - j15.29) \Omega$ (in rectangular form)

$I_{rms} = \frac{V_{rms}}{Z_{eq}} = \frac{12\sqrt{2}}{24.25 \angle -39.09^\circ} = 0.494 \angle 39.09^\circ \text{ A}$

or, $i = 0.7 \cos(5000t + 39.09^\circ) \text{ A}$

②

Given $i = 10 \cos 1000t$ A

i.e. $i = \frac{10}{\sqrt{2}} \angle 0^\circ$ A (rms)

$\omega = 1000$ rad/s

so, $X_C = \frac{1}{j\omega C} = -j10 \Omega$

$X_L = j\omega L = j5 \Omega$

apply nodal analysis at v_a

$$-i + \frac{v_a}{-j10} + \frac{v_a - v_b}{10} = 0 \quad \text{--- (1)}$$

apply nodal analysis at v_b

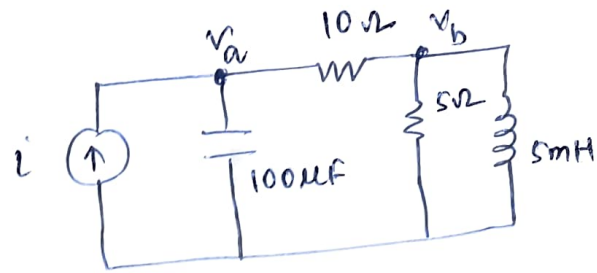
$$\frac{v_b}{5} + \frac{v_b}{j5} + \frac{v_b - v_a}{10} = 0 \quad \text{--- (2)}$$

so,
$$v_b = \frac{v_a}{3 - 2j} \quad \text{--- (3)}$$

put v_b from eqⁿ (3) into eqⁿ (1) & solve for v_a

$$v_a = 61.83 \angle -47.73^\circ \text{ V} \rightarrow \text{(rms)}$$

or,
$$v_a = 87.45 \cos(1000t - 47.73^\circ) \text{ V}$$

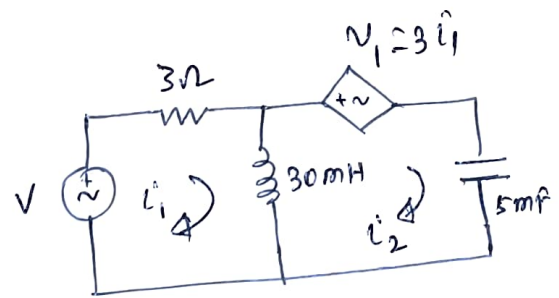


③

Given

$$v = 10\sqrt{2} \cos(\omega t + 45^\circ) \text{ V}$$

$$\omega = 100 \text{ rad/s}$$



so, $v = 10 \angle 45^\circ \text{ V (rms)}$

so, $x_c = \frac{1}{j\omega c} = -2j \Omega$, $x_L = j\omega L = j3 \Omega$

Apply KVL in loop 1

$$v = 3i_1 + 3j(i_1 - i_2) \quad \text{--- (1)}$$

KVL in loop 2 :- $3j(i_1 - i_2) = 3i_1 - 2ji_2 \quad \text{--- (2)}$

solve eqn (1) & get expression for i_2

$$i_2 = \frac{v - 3i_1 - 3ji_1}{-3j} \quad \text{--- (3)}$$

Put i_2 expression from eqn (3) into eqn (2) to get i_1

$$i_1 = \frac{-v}{3(2j-4)}$$

$$i_1 = 1.054 \cos(100t + 71.56^\circ) \text{ A}$$

4)

Given

$$i_s = 120\sqrt{2} \cos(1000t + 90^\circ) \text{ A}$$

$$\text{so, } \omega = 1000 \text{ rad/s}$$

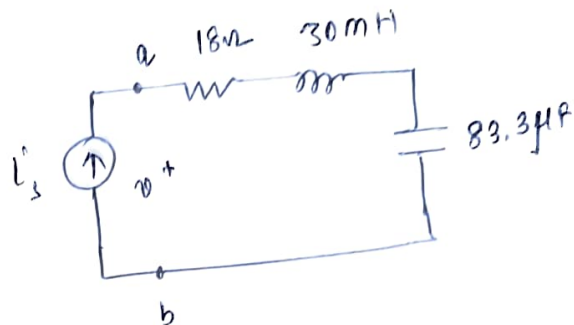
$$X_L = j\omega L = j30 \Omega$$

$$X_C = \frac{1}{j\omega C} = -j12 \Omega$$

$$v = i_s (R + X_L + X_C)$$

$$v = i_s (18 + j18)$$

$$v = 4320 \cos(1000t + 135^\circ) \text{ V}$$



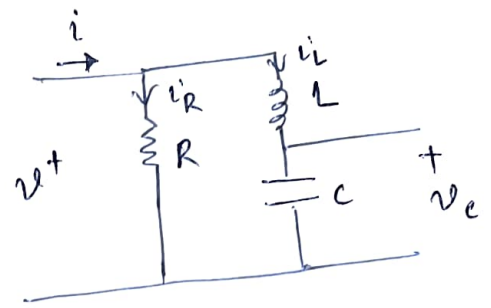
(5.)

Given $v = 5\sqrt{2} \cos 5t \text{ V}$

$R = 2 \Omega$

$L = 1 \text{ mH}$

$C = 0.06 \text{ F}$



so, $\omega = 5 \text{ rad/s}$

$X_L = j\omega L = j5 \Omega$; $X_C = \frac{1}{j\omega C} = -j3.33 \Omega$

so, $i_R = \frac{v}{R} = 2.5\sqrt{2} \cos 5t \text{ A}$

$i_L = \frac{v}{X_L + X_C} = 4.23 \cos(5t - 90^\circ) \text{ A}$

so, $i = i_R + i_L = 5.51 \cos(5t - 50.11^\circ) \text{ A}$

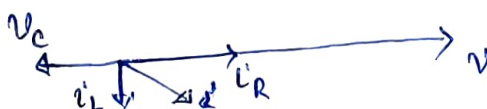
so, input impedance = $\frac{\text{input voltage}}{\text{input current}}$

$Z_{in} = \frac{5\sqrt{2} \angle 0^\circ}{5.51 \angle -50.11^\circ} \Omega$

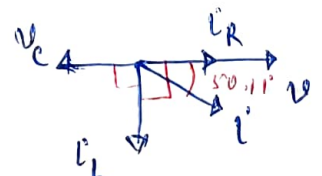
$Z_{in} = (0.82 + j0.98) \Omega$

or, $Z_{in} = 1.28 \angle 50.11^\circ \Omega$

Phasor :-



Phasor



- ⑥ Lighting Load : $10 \text{ kW} + j 0 \text{ kVAR}$
 resistance heating :- $200 \times 150 = 30 \text{ kW} = 30 \text{ kW} + j 0 \text{ kVAR}$
 Induction Furnace :- $360 \text{ kW} + j 400 \text{ kVAR}$

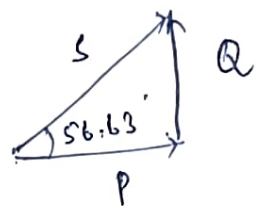
so, $S_{\text{total}} = (400 + j 400) \text{ kVA}$ (in rectangular form)

or, $565.685 \angle 45^\circ \text{ kVA}$ (in polar form)

- ⑦ Given $V = 200 \text{ V}$
 $f = 60 \text{ Hz}$
 $P = 10 \text{ kW}$
 $\cos \phi = 0.55 \text{ (lead)}$; $\phi = \cos^{-1}(0.55) = 56.63^\circ$

⑧ $P = VI \cos \phi$

so, $I = \frac{P}{V \cos \phi} = 90.91 \text{ A}$



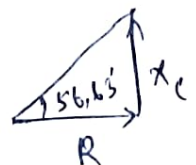
so, $\tan(56.63^\circ) = \frac{Q}{P}$

so, $Q = P \times \tan 56.63 = 15.18 \text{ kVAR}$

- ⑨ Since the power factor is leading (not exactly 90°),
 R - C ^{series} combination of load will be there

so, $R = \frac{P}{I^2} = \frac{10000}{(90.91)^2} = 1.21 \Omega$

$\tan(56.63^\circ) = \frac{X_C}{R}$



so, $X_C = 1.837 \Omega$

$X_C = \frac{1}{j\omega C}$ & $\omega = 2\pi \times 60 \text{ rad/s}$; so, $C = 1.44 \text{ mF}$

8

Wattmeter reading = 20 W

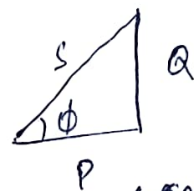
Ammeter reading = 2 A (rms)

so, $P = I^2 R$

$$R = \frac{P}{I^2} = \frac{20}{2^2} = 5 \Omega$$

Now, complex power (S) = $V_{rms} \times I_{rms}$

$$S = 26 \times 2 = 52 \text{ VA}$$



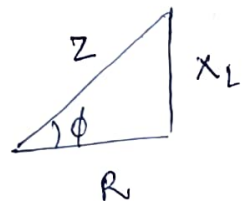
(Complex power triangle)

so, from complex power triangle

$$\cos \phi = \frac{P}{S} = \frac{20}{52}$$

$$\sin \phi = \frac{Q}{S} = \frac{\sqrt{S^2 - P^2}}{S} = \frac{48}{52}$$

$$\tan \phi = \frac{Q}{P} = \frac{48}{20}$$



so, we know that we can

compare Impedance Triangle &

complex power triangle because both are equivalent in terms of ϕ

so, $\tan \phi = \frac{X_L}{R}$

$$\frac{48}{20} = \frac{X_L}{R} \Rightarrow X_L = 12 \Omega$$

$$\omega L = 12$$

$$2\pi \times 60 \times L = 12$$

$$\Rightarrow L = 0.0318 \text{ H}$$

⑨ Given $S = 10 \text{ kVA}$

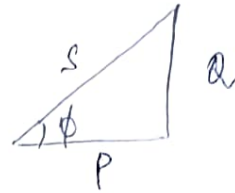
$$\cos \phi = 0.7 \text{ (lag)}$$

$$P = S \cos \phi$$

$$P = 10 \times 0.7 = 7 \text{ kW}$$

$$Q = S \sin \phi = 7.14 \text{ kVAR}$$

$$P_{\text{net}} = (7 + 20) \text{ kW} = 27 \text{ kW}$$



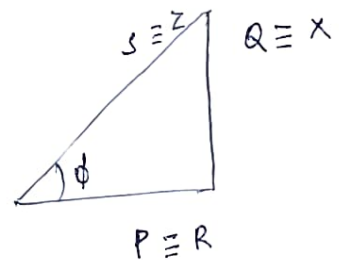
$$\text{plant power factor } (\cos \phi) = \frac{P_{\text{net}}}{S}$$

$$\cos \phi = \frac{P_{\text{net}}}{\sqrt{Q^2 + P_{\text{net}}^2}} = \frac{27}{\sqrt{(7.14)^2 + 27^2}}$$

$$\boxed{\cos \phi = 0.967 \text{ (lag)}}$$

$$\text{Total kVA (S)} = \sqrt{P_{\text{net}}^2 + Q^2} = \sqrt{27^2 + 7.14^2} = 27.93 \text{ kVA}$$

⑩ Given $\cos \phi = 0.85$
so, $\phi = 31.79^\circ \text{ (lag)}$



$$\tan \phi = \frac{Q_{\text{net}}}{P_{\text{net}}}$$

$$\tan(31.79) = \frac{Q_{\text{net}}}{27}$$

$$Q_{\text{net}} = 16.74 \text{ kVAR}$$

$$Q_{\text{net}} = Q_L - Q_C$$

$$16.74 = 7.14 - Q_C$$

$$Q_C = -9.6 \text{ kVAR} \text{ (-ve sign shows supply of reactive power)}$$

$$\text{so, } \boxed{Q_C = 9.6 \text{ kVAR}} \text{ Ans.}$$